

Meridian Newsroom — Local Setup Guide

Headless Drupal 11 + Next.js 14 article page. ~10–45 min from a fresh clone.

What you're getting

A two-half headless project. One Drupal 11 backend (content + JSON:API) and one Next.js 14 frontend rendering a single article page from that JSON:API. Both run locally; the dev URL is `http://localhost:3000/news/<slug>`.

Component	Tech	Runs at
Backend	Drupal 11, PHP 8.4, via DDEV (Docker)	<code>https://meridian.ddev.site</code>
Frontend	Next.js 14 App Router, TypeScript, RSC	<code>http://localhost:3000</code>

Prerequisites

Install all of these *before* cloning. On macOS via Homebrew:

```
brew install --cask orbstack          # or Docker Desktop – OrbStack is faster on Apple Silicon
brew install ddev mkcert node pnpm git
mkcert -install                       # installs the mkcert root CA into your system
trust store
```

Tool	Min version	Why
Docker (OrbStack or Docker Desktop)	—	DDEV's container runtime
DDEV	1.23+	Boots the Drupal stack (web + db) in containers
mkcert	any	Issues a local TLS cert DDEV uses for <code>*.ddev.site</code>
Node	18.18+ (20+ recommended)	Runs Next.js
pnpm	9+	Workspace package manager
git	any modern	Cloning

Windows users: use WSL2 + Docker Desktop with WSL backend. Run all commands inside the WSL2 shell, not PowerShell. The `NODE_EXTRA_CA_CERTS` step (below) is especially fiddly on Windows — the mkcert CA path differs.

Setup — 7 steps

Step 1. Clone and install JS deps.

```
git clone https://github.com/Jdouville8/headless-next-drupal-example.git meridian
cd meridian
pnpm install # installs apps/web deps via workspaces
```

Step 2. Boot the Drupal containers. **First boot will sudo-prompt** to add `meridian.ddev.site` to `/etc/hosts` — type your password when asked.

```
cd apps/drupal
ddev start # sudo prompt happens here, once
```

Step 3. Install Drupal core + contrib via Composer (runs inside the container).

```
ddev composer install # ~2 min on a fresh machine
```

Step 4. Install Drupal itself from the committed config, then seed the sample article.

```
ddev drush si --existing-config -y # builds schema from config/sync/
ddev drush scr scripts/seed-content.php # creates the demo article (nid=3)
```

Admin login if you need it: `ddev launch $(ddev drush uli)` — opens a one-shot login URL. Local creds are `admin / admin`.

Step 5. Configure the Next.js env (back in the repo root).

```
cd ../web
cat > .env.local <<'EOF'
NEXT_PUBLIC_USE MOCK_DATA=false
DRUPAL_BASE_URL=https://meridian.ddev.site
NEXT_PUBLIC_SHOW_DEV_PANEL=true
REVALIDATE_SECRET=replace-with-a-real-random-string
EOF

# then regenerate the secret with a real value:
SECRET=$(openssl rand -hex 32)
sed -i.bak "s|replace-with-a-real-random-string|$SECRET|" .env.local && rm .env.local.bak
```

Step 6 — the critical one. Export the mkcert CA path *in the same shell before* launching `pnpm dev`. Putting this in `.env.local` does *not* work — Node reads `NODE_EXTRA_CA_CERTS` during TLS init, before Next.js loads its env files.

```
export NODE_EXTRA_CA_CERTS="$(mkcert -CAROOT)/rootCA.pem"
```

If you skip this you'll see `UNABLE_TO_VERIFY_LEAF_SIGNATURE` when the Next.js server tries to fetch from Drupal. Same shell, every time you start the dev server.

Step 7. Start the dev server.

```
pnpm dev
# → ready at http://localhost:3000
# → article at http://localhost:3000/news/meridian-industrial-debuts-continental-
logistics-program-next-generation-fleet-operators
```

Verifying it works

Open the article URL above in a browser. You should see a full press-release page with an amber title band, a navy pull-quote band with a portrait, and a footer with the Meridian wordmark.

Sanity check from the terminal:

```
curl -s -o /dev/null -w '%{http_code}\n' \
  http://localhost:3000/news/meridian-industrial-debuts-continental-logistics-program-
next-generation-fleet-operators
# → 200
```

If you see **200** with the expected layout, you're done. Open `Article Page (standalone).html` from the repo root in a second tab at 1440 px wide to do a side-by-side visual compare against the design.

Common gotchas

"Cannot find module './352.js'" in the browser, after running `pnpm build`.

The dev server and `pnpm build` both write to `apps/web/.next/`. Running build while dev is up clobbers the dev server's webpack chunks. Fix:

```
pkill -f 'next dev'
rm -rf apps/web/.next
cd apps/web
export NODE_EXTRA_CA_CERTS="$(mkcert -CAROOT)/rootCA.pem"
pnpm dev
```

Rule: stop dev before running build.

UNABLE_TO_VERIFY_LEAF_SIGNATURE in the Next.js logs.

You forgot Step 6 (or you're in a fresh shell). Re-export `NODE_EXTRA_CA_CERTS` and relaunch `pnpm dev`. Consider adding the export line to your shell rc (`~/.zshrc`) if you'll do this often.

DDEV port conflicts on `ddev start`.

Another DDEV project (or a host Apache/Caddy) holds 80/443. Run `ddev poweroff` to stop everything, then `cd apps/drupal && ddev start` for just this one. If you keep multiple DDEV projects running, DDEV's router multiplexes them by hostname — that usually just works.

JSON:API returns 500 with 'path' not found when the Next.js server fetches.
The custom `meridian_path_filter` module is disabled. It's required because Drupal core can't filter nodes by `path.alias` via JSON:API. Re-enable:

```
cd apps/drupal && ddev drush en meridian_path_filter -y
```

Drupal content change isn't showing. Hard-refresh the browser (⌘⇧R). If it still doesn't, clear all caches:

```
cd apps/drupal && ddev drush cr           # Drupal cache
pkill -f 'next dev'                      # dev server (image-optimizer cache)
rm -rf apps/web/.next
# restart dev with the NODE_EXTRA_CA_CERTS export
```

What you can do once it's running

- **Edit content in Drupal:** log in via `ddev launch $(ddev drush uli)` (run from `apps/drupal/`). Edit the article, save, hard-refresh the Next.js page.
- **Test the dev panel** — bottom-right floating box. Toggle *Show JSON mapping* to label every DOM region with its source field. Switch the accent (amber → emerald → cobalt → scarlet) to verify the design tokens reflow.
- **Trigger ISR revalidation** from a script:

```
SECRET=$(grep ^REVALIDATE_SECRET= apps/web/.env.local | cut -d= -f2)
curl -X POST "http://localhost:3000/api/revalidate?secret=$SECRET&path=/news/<slug>"
```

Where to look when something goes deeper

For...	Read
Full reference + deploy stubs + edge cases	<code>RUNBOOK.md</code> at the repo root
Drupal-side specifics, fields, deviations from the spec	<code>apps/drupal/HANDOFF.md</code>
The data contract between Drupal and Next.js	<code>JSON_CONTRACT.md</code> at the repo root
How a new paragraph type gets added	<code>RUNBOOK.md</code> § "How to add a new paragraph type"
Visual-parity target	<code>Article Page (standalone).html</code> at the repo root — open in a browser